

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

Department of Artificial Intelligence and Machine Learning

CSA1012 – Software Engineering

SOFTWARE REQUIREMENT SPECIFICATION

AgriTwin

Digital Twin Platform for Precision Farming and Crop Health Prediction

Scenario-Based Requirement Analysis

Paleru Dharani Govardhan (192525280)

Course Code: CSA1012

Course Name: Software Engineering

Programme: B.E. / B.Tech – Artificial Intelligence and Machine Learning

Table of Contents

Table of Contents	2
1. Introduction	7
1.1 Purpose	7
1.2 Scope	7
1.3 Definitions	7
1.4 Abbreviations	8
1.5 Overview	9
2. Problem Description	10
2.1 Current System	10
2.2 Existing Problems	10
2.3 Need for the Proposed System	10
2.4 Objectives	11
2.5 Business Domain	11
3. Stakeholder Analysis	12
3.1 Farmer	12
3.2 Agricultural Expert	12
3.3 Government Agriculture Officer	12
3.4 Farm Manager	12
3.5 IoT Sensor System	13
3.6 Drone System	13
3.7 Weather Service	13
3.8 Satellite Data Provider	13
3.9 Administrator	14
3.10 AI Prediction Engine	14
4. Scope of the System	15
4.1 In Scope	15
4.2 Out of Scope	15
4.3 Future Scope	15
5. Functional Requirements	17
FR-01 – User Registration	17
FR-02 – User Authentication	17
FR-03 – Field Registration	17

FR-04 – Crop Profile Configuration	17
FR-05 – IoT Sensor Data Ingestion	18
FR-06 – Drone Imagery Ingestion	18
FR-07 – Weather Data Retrieval	18
FR-08 – Satellite Imagery Retrieval	18
FR-09 – Digital Twin Synchronization.....	19
FR-10 – Crop Health Prediction	19
FR-11 – Disease Detection	19
FR-12 – Pest Outbreak Prediction.....	19
FR-13 – Irrigation Recommendation	20
FR-14 – Fertilization Recommendation	20
FR-15 – Pest Management Recommendation	20
FR-16 – Alert Generation.....	20
FR-17 – Notification Delivery	21
FR-18 – Digital Twin Dashboard	21
FR-19 – Crop Growth Monitoring.....	21
FR-20 – Field Simulation	21
FR-21 – Predictive Report Generation	22
FR-22 – Expert Consultation Request	22
FR-23 – Expert Response Submission	22
FR-24 – Regional Crop Statistics Access	22
FR-25 – Advisory Publication.....	23
FR-26 – User Account Management.....	23
FR-27 – System Configuration Management	23
FR-28 – System Log Monitoring	23
FR-29 – Multi-Field Management	24
6. Non-Functional Requirements	25
6.1 Performance.....	25
6.2 Security.....	25
6.3 Reliability	25
6.4 Availability.....	25
6.5 Scalability	25
6.6 Maintainability	25
6.7 Usability	26
6.8 Portability	26

6.9 Interoperability	26
6.10 Data Integrity.....	26
6.11 Backup	26
6.12 Recovery.....	26
6.13 Response Time	26
6.14 Compliance.....	26
6.15 Accessibility.....	27
6.16 Summary Table	27
7. Actors.....	28
7.1 Primary Actors.....	28
7.2 Secondary Actors.....	28
8. Use Cases	29
UC-01 – Register and Login	29
UC-02 – Register a New Field.....	29
UC-03 – Configure Crop Profile	29
UC-04 – View Digital Twin Dashboard	30
UC-05 – View Crop Health Prediction.....	30
UC-06 – Receive Alert Notification	31
UC-07 – Request Irrigation Recommendation	31
UC-08 – Request Fertilization Recommendation	31
UC-09 – View Pest Outbreak Prediction	32
UC-10 – Run Field Simulation	32
UC-11 – Generate Predictive Report.....	32
UC-12 – Request Expert Consultation	33
UC-13 – Respond to Consultation.....	33
UC-14 – View Regional Crop Statistics.....	34
UC-15 – Publish Regional Advisory	34
UC-16 – Manage User Accounts.....	34
UC-17 – Monitor System Logs.....	35
UC-18 – Ingest Automated Sensor Data	35
9. UML Use Case Diagram.....	36
10. Data Requirements.....	37
10.1 Entities and Attributes	37
10.2 Relationships	38
10.3 Entity Relationship Diagram.....	39

11. External Interface Requirements	40
11.1 User Interface.....	40
11.2 Hardware Interface.....	40
11.3 Software Interface	40
11.4 Communication Interface	40
11.5 API Interface	40
11.6 Cloud Services	40
12. System Features	42
12.1 Digital Twin Field Modelling	42
12.2 Multi-Source Real-Time Data Integration	42
12.3 AI-Driven Crop Health Prediction.....	42
12.4 Disease Detection from Imagery	42
12.5 Pest Outbreak Forecasting	42
12.6 Intelligent Irrigation Recommendations	42
12.7 Fertilization Guidance	42
12.8 Pest Management Recommendations.....	43
12.9 Real-Time Alerting and Notifications.....	43
12.10 Visualization Dashboards	43
12.11 Crop Growth Timeline Monitoring	43
12.12 Field Simulation ("What-If" Analysis)	43
12.13 Predictive and Historical Reporting	43
12.14 Expert Consultation Workflow.....	43
12.15 Regional Aggregated Analytics	43
12.16 Role-Based Access Control and Administration	44
13. Assumptions.....	45
14. Constraints.....	46
15. Requirement Validation	47
15.1 Completeness.....	47
15.2 Consistency.....	47
15.3 Correctness.....	47
15.4 Traceability.....	47
15.5 Ambiguity	47
15.6 Redundancy.....	48
15.7 Missing Requirements	48
15.8 Suggested Improvements	48

15.9 Validation Checklist	48
16. Conclusion.....	50
Appendix A: Supplementary Diagrams	51
A.1 Context Diagram (DFD Level 0).....	51
A.2 Level-0 Data Flow Diagram	51
A.3 System Architecture Diagram	52
A.4 Component Diagram.....	53
A.5 Deployment Diagram	54

1. Introduction

1.1 Purpose

The purpose of this Software Requirement Specification (SRS) document is to define, in a complete and unambiguous manner, the functional and non-functional requirements of AgriTwin, a Digital Twin platform designed for precision farming and crop health prediction. This document is intended to serve as the primary reference for the design, development, testing, and evaluation of the AgriTwin system. It describes the intended behaviour of the system from the perspective of all stakeholders, including farmers, agricultural experts, government agencies, and system administrators, and establishes a shared understanding between the project team and its evaluators regarding what the system will and will not do.

This SRS has been prepared as part of the CSA1012 – Software Engineering course assignment, following the scenario-based requirement analysis methodology, and adheres to the general structure recommended by IEEE Std 830 for software requirement specifications.

1.2 Scope

AgriTwin is a cloud-based Digital Twin platform that continuously synchronizes real-time and historical agricultural field data collected from IoT sensors, drones, weather stations, and satellite imagery to construct and maintain a virtual replica of a physical farm. The platform applies Artificial Intelligence and Machine Learning models to this synchronized data to predict crop health conditions, detect diseases and pest outbreaks at an early stage, and recommend optimized irrigation, fertilization, and pest management practices.

The system provides role-based dashboards and visualization tools for farmers, farm managers, agricultural experts, and government agriculture officers, enabling proactive, data-driven farm management. AgriTwin is intended to improve crop productivity, reduce resource wastage, and promote sustainable agricultural practices at both the individual farm level and the regional level.

1.3 Definitions

Term	Definition
Digital Twin	A dynamic virtual representation of a physical farm field that is continuously synchronized with real-world sensor and imagery data to mirror the current and predicted state of the field.

Term	Definition
Precision Farming	A farm management approach that uses data-driven technologies to observe, measure, and respond to inter- and intra-field variability in crops.
Crop Health Prediction	The use of AI/ML models to forecast the physiological condition of a crop, including stress, disease, or nutrient deficiency, before visible symptoms fully develop.
NDVI	Normalized Difference Vegetation Index; a numerical indicator derived from imagery used to assess live green vegetation and plant health.
IoT Sensor	An internet-connected physical device deployed in the field to measure parameters such as soil moisture, temperature, and humidity.
Recommendation Engine	A system module that converts prediction outputs into actionable guidance for irrigation, fertilization, or pest management.
Stakeholder	Any individual, organization, or system that has an interest in, or is affected by, the AgriTwin platform.

1.4 Abbreviations

Abbreviation	Expansion
SRS	Software Requirement Specification
IoT	Internet of Things
AI	Artificial Intelligence
ML	Machine Learning
UI	User Interface
UX	User Experience
API	Application Programming Interface
DB	Database
ERD	Entity Relationship Diagram
DFD	Data Flow Diagram
UML	Unified Modeling Language
NDVI	Normalized Difference Vegetation Index
FR	Functional Requirement
NFR	Non-Functional Requirement
IAM	Identity and Access Management

Abbreviation	Expansion
TLS	Transport Layer Security
GPS	Global Positioning System
SLA	Service Level Agreement

1.5 Overview

The remainder of this document is organized as follows. Section 2 describes the current farming practices, the problems they present, and the need for the proposed system. Section 3 identifies and analyses all stakeholders of AgriTwin. Section 4 defines the scope of the system in terms of what is included, excluded, and reserved for future enhancement. Section 5 enumerates the functional requirements, and Section 6 specifies the non-functional requirements with measurable targets. Section 7 lists the primary and secondary actors, while Section 8 documents the use cases derived from stakeholder interactions. Section 9 presents the UML Use Case Diagram. Section 10 details the data requirements and Entity Relationship Diagram. Section 11 specifies external interface requirements, and Section 12 elaborates the major system features. Sections 13 and 14 record the assumptions and constraints respectively. Section 15 presents the requirement validation process and checklist, and Section 16 concludes the document.

2. Problem Description

2.1 Current System

At present, the majority of farmers rely on manual field inspection, visual observation, and traditional farming knowledge passed down through experience to assess crop condition, soil quality, and irrigation needs. Field visits are conducted periodically, often at intervals that are too infrequent to catch early-stage stress indicators. Where technology is used, it typically takes the form of isolated point solutions, such as a standalone soil moisture sensor application or a separate weather forecasting app, none of which are integrated with one another or with a unified view of the farm.

Existing precision farming tools generally provide raw sensor readings or satellite imagery without contextual interpretation, and they rarely offer predictive capability. Farmers must manually correlate data from multiple disconnected sources and rely on their own judgment or delayed consultation with agricultural experts to decide on interventions such as irrigation scheduling, fertilizer application, or pest control measures.

2.2 Existing Problems

- Manual field inspections are time-consuming, infrequent, and prone to human error, often failing to detect early-stage crop stress, disease, or pest infestation.
- Existing precision farming tools provide isolated, single-purpose monitoring features (for example, moisture-only or weather-only applications) without an integrated virtual model of the entire farm.
- Delayed detection of diseases, pest infestations, irrigation problems, and nutrient deficiencies leads to avoidable yield losses and increased operational costs.
- Agricultural data originates from multiple fragmented sources (sensors, drones, weather services, satellite providers), making it difficult for farmers to form a single, coherent picture of field health.
- There is no mechanism for simulating the outcome of a proposed intervention (for example, a change in irrigation schedule) before it is physically applied in the field.
- Farmers in remote areas often lack timely access to agricultural experts, delaying critical decisions.
- Government agencies lack a consolidated, data-driven view of regional crop health for policy planning and food security monitoring.

2.3 Need for the Proposed System

There is a clear need for an intelligent Digital Twin platform that continuously synchronizes real-time farm data from multiple heterogeneous sources into a single virtual model of

the field, applies AI-driven predictive analysis to detect crop health issues before they become visually apparent, and recommends optimized, data-backed farming practices. Such a system would enable proactive rather than reactive farm management, reduce the dependency on manual inspection, and support both individual farmers and government agencies in achieving better agricultural outcomes.

2.4 Objectives

- Develop a Digital Twin platform that creates and maintains a virtual representation of agricultural fields.
- Collect real-time data from IoT sensors, drones, weather stations, and satellite imagery.
- Predict crop health, diseases, and pest outbreaks using Artificial Intelligence and Machine Learning models.
- Monitor soil moisture, nutrient levels, irrigation status, and environmental conditions continuously.
- Recommend irrigation schedules, fertilization plans, and pest management actions based on predictive analysis.
- Provide visualization dashboards and crop growth monitoring tools for all user roles.
- Generate predictive reports for farmers, experts, and government agencies.
- Improve resource utilization (water, fertilizer, labour) while promoting sustainable farming practices.

2.5 Business Domain

AgriTwin operates within the precision agriculture and agri-technology domain, at the intersection of IoT, remote sensing, cloud computing, and applied artificial intelligence. The business domain encompasses individual and cooperative farming operations, agribusiness advisory services, and public-sector agricultural extension programmes. The platform is designed to be commercially viable as a subscription-based Software-as-a-Service (SaaS) offering for individual farmers and farm management companies, while also supporting bulk or institutional licensing for government agricultural departments seeking regional crop-health visibility.

3. Stakeholder Analysis

This section identifies every stakeholder of the AgriTwin platform, including human roles and external systems that interact with it, along with their responsibilities and mode of interaction with the system.

3.1 Farmer

Role: Primary end-user and owner or operator of the agricultural field being monitored.

Responsibilities: Registers fields and crop details, reviews the Digital Twin dashboard, responds to alerts and recommendations, and applies recommended interventions such as irrigation or fertilization.

Interaction with System: Uses the mobile and web application to view field status, crop health predictions, and recommendations; receives push notifications and alerts; requests consultations with agricultural experts.

3.2 Agricultural Expert

Role: Domain specialist who provides advisory support and validates AI-generated recommendations.

Responsibilities: Reviews flagged cases where AI confidence is low, responds to farmer consultation requests, and issues professional guidance on disease and pest management.

Interaction with System: Accesses the advisory portal to view field data, prediction outputs, and farmer queries; submits expert responses and recommendations through the system.

3.3 Government Agriculture Officer

Role: Public-sector stakeholder responsible for regional agricultural policy and food security monitoring.

Responsibilities: Monitors aggregated, anonymized regional crop health statistics, issues public advisories during disease or pest outbreaks, and uses platform data for planning subsidy and support programmes.

Interaction with System: Accesses read-only regional dashboards and exports compliance and advisory reports; does not interact with individual private farm data without authorization.

3.4 Farm Manager

Role: Operational manager responsible for overseeing one or more fields on behalf of a farm owner or cooperative.

Responsibilities: Configures field and crop profiles, monitors multiple fields simultaneously, coordinates labour and resource allocation based on system recommendations.

Interaction with System: Uses a management dashboard to view multi-field summaries, assign tasks, and track the status of recommended interventions across fields.

3.5 IoT Sensor System

Role: Automated external system providing continuous field-level environmental data.

Responsibilities: Measures and transmits soil moisture, temperature, humidity, and related parameters at regular intervals.

Interaction with System: Sends sensor readings to the platform via the IoT gateway using lightweight messaging protocols; does not consume system output directly.

3.6 Drone System

Role: Automated aerial data-capture system providing high-resolution field imagery.

Responsibilities: Captures multispectral and RGB imagery of fields on a scheduled or on-demand basis for vegetation index computation.

Interaction with System: Uploads captured imagery and metadata to the platform through the data ingestion layer for processing by the Digital Twin engine.

3.7 Weather Service

Role: External third-party data provider supplying meteorological information.

Responsibilities: Provides current conditions and forecast data relevant to the geographic location of each registered field.

Interaction with System: Exposes a REST API that the AgriTwin data ingestion layer queries periodically to retrieve weather and forecast data.

3.8 Satellite Data Provider

Role: External third-party provider of satellite-based remote sensing imagery.

Responsibilities: Supplies periodic satellite imagery and derived indices (such as NDVI) covering registered field boundaries.

Interaction with System: Provides imagery through an API or scheduled data feed consumed by the data ingestion and integration layer.

3.9 Administrator

Role: System administrator responsible for platform operation, configuration, and security.

Responsibilities: Manages user accounts and access permissions, configures system-wide settings, monitors system logs and performance, and manages integration credentials for external data providers.

Interaction with System: Uses the administrative console to manage users, monitor system health, configure thresholds, and review audit logs.

3.10 AI Prediction Engine

Role: Internal automated subsystem responsible for analytical processing.

Responsibilities: Continuously analyses synchronized field data to generate crop health, disease, and pest outbreak predictions, and feeds these outputs to the recommendation engine.

Interaction with System: Operates as an internal service that consumes data from the Digital Twin engine and produces prediction records consumed by the recommendation and alerting components.

4. Scope of the System

4.1 In Scope

- Creation and continuous synchronization of a Digital Twin virtual model for each registered field.
- Ingestion of data from IoT soil and climate sensors, drone imagery, weather station APIs, and satellite imagery providers.
- AI/ML-based prediction of crop health status, plant disease presence, and pest outbreak likelihood.
- Generation of irrigation, fertilization, and pest management recommendations based on prediction outputs.
- Role-based dashboards and visualization tools for farmers, farm managers, agricultural experts, government officers, and administrators.
- Alert and notification generation for critical field conditions.
- Predictive and historical report generation in exportable formats.
- Field and crop growth simulation ("what-if" scenario analysis) using the Digital Twin model.
- User authentication, role-based access control, and administrative account management.
- Regional, aggregated crop-health analytics for government agriculture officers.

4.2 Out of Scope

- Direct control of physical farm machinery or irrigation actuators (the system recommends actions; it does not autonomously operate physical equipment in the initial release).
- Financial transaction processing, such as crop marketplace sales, loans, or insurance claim processing.
- Manufacture, sale, or physical maintenance of IoT sensors, drones, or other hardware devices.
- Livestock monitoring or management (the system is limited to crop and field monitoring).
- Direct integration with national land-registry or property-title systems.

4.3 Future Scope

- Integration with automated irrigation actuators and variable-rate fertilizer applicators for closed-loop control.

- Blockchain-based traceability of crop produce from field to market.
- Marketplace integration enabling farmers to sell produce directly through the platform.
- Expansion of AI models to support additional crop varieties and region-specific disease libraries.
- Integration with agricultural insurance and micro-credit platforms using verified Digital Twin data.
- Support for multilingual voice-based interaction for low-literacy users.

5. Functional Requirements

This section specifies the functional requirements of the AgriTwin system. A total of twenty-nine (29) functional requirements have been identified, each uniquely numbered, prioritized, and described in terms of its inputs and outputs, exceeding the minimum requirement of twenty-five.

FR-01 – User Registration

Field	Detail
Description	The system shall allow Farmers, Farm Managers, and Agricultural Experts to register an account by providing personal, contact, and role-specific details.
Priority	High
Inputs	Name, email, phone number, password, role selection
Outputs	Confirmed user account; verification email

FR-02 – User Authentication

Field	Detail
Description	The system shall authenticate registered users via email/username and password before granting access to role-specific features.
Priority	High
Inputs	Username or email, password
Outputs	Authenticated session token; access granted or denied

FR-03 – Field Registration

Field	Detail
Description	The system shall allow Farmers and Farm Managers to register a new field by defining its geographic boundary, area, and soil type.
Priority	High
Inputs	Field name, GPS boundary polygon, area, soil type
Outputs	Newly created field record linked to a Digital Twin instance

FR-04 – Crop Profile Configuration

Field	Detail
Description	The system shall allow users to assign a crop type and variety to a registered field, including planting date and expected growth duration.
Priority	High
Inputs	Crop name, variety, planting date
Outputs	Configured crop profile associated with the field

FR-05 – IoT Sensor Data Ingestion

Field	Detail
Description	The system shall automatically ingest soil moisture, temperature, and humidity readings transmitted by registered IoT sensors at defined intervals.
Priority	High
Inputs	Sensor ID, parameter values, timestamp
Outputs	Stored sensor reading records; updated field state

FR-06 – Drone Imagery Ingestion

Field	Detail
Description	The system shall accept uploaded drone-captured imagery and associated metadata for a given field and processing timestamp.
Priority	High
Inputs	Image file, capture time, drone ID, field ID
Outputs	Stored imagery record; computed NDVI index

FR-07 – Weather Data Retrieval

Field	Detail
Description	The system shall periodically retrieve current and forecast weather data from the configured Weather Service API for each registered field location.
Priority	High
Inputs	Field GPS coordinates
Outputs	Stored weather data record (temperature, humidity, rainfall, wind)

FR-08 – Satellite Imagery Retrieval

Field	Detail
Description	The system shall periodically retrieve satellite imagery covering registered field boundaries from the configured Satellite Data Provider.
Priority	Medium
Inputs	Field boundary coordinates, date range
Outputs	Stored satellite image record with spectral band data

FR-09 – Digital Twin Synchronization

Field	Detail
Description	The system shall synchronize the virtual Digital Twin model of each field with the latest ingested sensor, imagery, and weather data.
Priority	High
Inputs	Latest sensor, imagery, and weather records
Outputs	Updated Digital Twin model state; synchronization timestamp

FR-10 – Crop Health Prediction

Field	Detail
Description	The system shall apply AI/ML models to synchronized field data to generate a crop health status prediction with a confidence score.
Priority	High
Inputs	Synchronized field data (moisture, NDVI, weather)
Outputs	Crop health prediction record with confidence score

FR-11 – Disease Detection

Field	Detail
Description	The system shall analyze drone and satellite imagery using AI models to detect visual indicators of plant disease.
Priority	High
Inputs	Processed imagery, historical disease patterns
Outputs	Disease detection prediction with affected area and confidence score

FR-12 – Pest Outbreak Prediction

Field	Detail
Description	The system shall predict the likelihood of pest outbreaks based on environmental conditions, historical patterns, and regional data.
Priority	High
Inputs	Weather data, historical pest incidence, field location
Outputs	Pest outbreak risk prediction with severity level

FR-13 – Irrigation Recommendation

Field	Detail
Description	The system shall generate irrigation recommendations, including timing and volume, based on soil moisture readings and weather forecasts.
Priority	High
Inputs	Soil moisture data, weather forecast, crop water requirement
Outputs	Irrigation recommendation record (schedule, volume)

FR-14 – Fertilization Recommendation

Field	Detail
Description	The system shall generate fertilization recommendations based on detected nutrient deficiency indicators and crop growth stage.
Priority	High
Inputs	Nutrient sensor data, crop growth stage
Outputs	Fertilization recommendation record (nutrient type, quantity, timing)

FR-15 – Pest Management Recommendation

Field	Detail
Description	The system shall recommend appropriate pest management actions when a pest outbreak risk is predicted above a configurable threshold.
Priority	High
Inputs	Pest outbreak prediction, crop type
Outputs	Pest management recommendation record

FR-16 – Alert Generation

Field	Detail
Description	The system shall automatically generate an alert when a prediction indicates a critical condition, such as high disease probability or severe water stress.
Priority	High
Inputs	Prediction results, configured alert thresholds
Outputs	Alert record with severity level and message

FR-17 – Notification Delivery

Field	Detail
Description	The system shall deliver generated alerts to the relevant user via in-app notification, and optionally SMS or email based on user preference.
Priority	High
Inputs	Alert record, user notification preferences
Outputs	Delivered notification; delivery status log

FR-18 – Digital Twin Dashboard

Field	Detail
Description	The system shall present a visual dashboard displaying the current state of the Digital Twin, including field map, sensor readings, and health status.
Priority	High
Inputs	Field ID, latest synchronized data
Outputs	Rendered dashboard view

FR-19 – Crop Growth Monitoring

Field	Detail
Description	The system shall display a timeline view of crop growth stages and corresponding health indicators over the crop cycle.
Priority	Medium
Inputs	Historical prediction and sensor data for the field
Outputs	Crop growth timeline visualization

FR-20 – Field Simulation

Field	Detail
Description	The system shall allow authorized users to run "what-if" simulations on the Digital Twin model to estimate the impact of a proposed intervention.
Priority	Medium
Inputs	Simulation parameters (e.g., proposed irrigation change)
Outputs	Simulated outcome report

FR-21 – Predictive Report Generation

Field	Detail
Description	The system shall generate downloadable predictive reports summarizing crop health, risk factors, and recommendations for a selected period.
Priority	Medium
Inputs	Field ID, report period, report type
Outputs	Generated report file (PDF)

FR-22 – Expert Consultation Request

Field	Detail
Description	The system shall allow a Farmer to submit a consultation query, optionally attaching field data, to an available Agricultural Expert.
Priority	Medium
Inputs	Farmer query text, field reference, optional images
Outputs	Created consultation record routed to an expert

FR-23 – Expert Response Submission

Field	Detail
Description	The system shall allow an Agricultural Expert to view assigned consultation queries and submit a written response.
Priority	Medium
Inputs	Consultation ID, expert response text
Outputs	Updated consultation record with expert response; notification to farmer

FR-24 – Regional Crop Statistics Access

Field	Detail
Description	The system shall provide Government Agriculture Officers with access to aggregated, anonymized regional crop health statistics.
Priority	Medium
Inputs	Region identifier, date range
Outputs	Aggregated regional statistics view/report

FR-25 – Advisory Publication

Field	Detail
Description	The system shall allow Government Agriculture Officers to publish regional advisories that are visible to all farmers within the affected region.
Priority	Medium
Inputs	Advisory text, affected region, validity period
Outputs	Published advisory visible on farmer dashboards

FR-26 – User Account Management

Field	Detail
Description	The system shall allow an Administrator to create, modify, deactivate, or delete user accounts and assign roles.
Priority	High
Inputs	User ID, action type, role assignment
Outputs	Updated user account record

FR-27 – System Configuration Management

Field	Detail
Description	The system shall allow an Administrator to configure system-wide parameters, including alert thresholds and data source credentials.
Priority	Medium
Inputs	Configuration parameter name and value
Outputs	Updated system configuration

FR-28 – System Log Monitoring

Field	Detail
Description	The system shall allow an Administrator to view system activity logs, including data ingestion status and prediction job execution status.
Priority	Medium
Inputs	Log query filters (date range, module)
Outputs	Filtered log entries displayed to administrator

FR-29 – Multi-Field Management

Field	Detail
Description	The system shall allow a Farm Manager to view and manage a consolidated summary of multiple fields under their supervision.
Priority	Medium
Inputs	Farm Manager ID
Outputs	Consolidated multi-field summary dashboard

6. Non-Functional Requirements

6.1 Performance

The system shall process and reflect newly ingested IoT sensor data in the Digital Twin dashboard within 30 seconds under normal load conditions. AI-based crop health predictions shall be generated within 5 minutes of the completion of a data synchronization cycle. The dashboard shall load within 3 seconds on a standard 4G mobile connection.

6.2 Security

All data transmitted between client applications, IoT devices, and the server shall be encrypted using TLS 1.2 or higher. User passwords shall be stored using a salted cryptographic hash (bcrypt or equivalent). The system shall enforce role-based access control (RBAC) such that each user can only access data and functions permitted for their assigned role.

6.3 Reliability

The system shall achieve a Mean Time Between Failures (MTBF) of at least 720 hours for core prediction and dashboard services. Failed data ingestion attempts from IoT sensors, drones, or external APIs shall be automatically retried up to 3 times before being logged as a failure.

6.4 Availability

The AgriTwin platform shall maintain a minimum uptime of 99.5% measured on a monthly basis, excluding scheduled maintenance windows communicated at least 24 hours in advance.

6.5 Scalability

The system architecture shall support horizontal scaling to accommodate at least a 10x increase in registered fields and concurrent users without requiring architectural redesign, through container-based deployment and load-balanced service instances.

6.6 Maintainability

The system shall be developed using a modular, service-oriented architecture such that individual components (data ingestion, AI prediction, recommendation engine) can be independently updated or replaced with less than 4 hours of scheduled downtime per component update.

6.7 Usability

The user interface shall be designed such that a first-time farmer user with basic smartphone literacy can complete field registration within 10 minutes without external assistance. The interface shall support at least two languages, including English and one regional language, in the initial release.

6.8 Portability

The client application shall be accessible via modern web browsers (Chrome, Firefox, Edge, Safari — latest two major versions) and shall be available as a native or hybrid mobile application for Android 9.0 and above.

6.9 Interoperability

The system shall expose a documented REST API supporting JSON payloads to enable integration with third-party weather services, satellite data providers, and future government agricultural information systems.

6.10 Data Integrity

The system shall validate all incoming sensor and imagery data against defined range and format constraints before persisting it, rejecting and logging any record that fails validation, to ensure that at least 99.9% of stored records are within expected value ranges.

6.11 Backup

The system shall perform automated incremental backups of the primary database every 6 hours and a full backup once every 24 hours, with backups retained for a minimum of 30 days.

6.12 Recovery

In the event of a system failure, the platform shall support a Recovery Time Objective (RTO) of 4 hours and a Recovery Point Objective (RPO) of 1 hour.

6.13 Response Time

Interactive user requests, such as viewing a dashboard or submitting a consultation query, shall receive a system response within 2 seconds under normal load (up to 1,000 concurrent users) for 95% of requests.

6.14 Compliance

The system shall comply with applicable regional data protection regulations governing the collection and storage of personal and location data, and shall provide mechanisms for user data export and account deletion upon request.

6.15 Accessibility

The web-based interface shall conform to WCAG 2.1 Level AA guidelines where feasible, including adequate colour contrast, keyboard navigability, and screen-reader-compatible labelling of interactive elements.

6.16 Summary Table

NFR Category	Measurable Target	Priority
Performance	Dashboard update $\leq 30s$; prediction ≤ 5 min	High
Security	TLS 1.2+; hashed passwords; RBAC	High
Reliability	MTBF ≥ 720 hours	High
Availability	$\geq 99.5\%$ monthly uptime	High
Scalability	Support 10x growth via horizontal scaling	Medium
Maintainability	< 4 hrs downtime per component update	Medium
Usability	Field registration completed in ≤ 10 min	High
Portability	Latest 2 browser versions; Android 9.0+	Medium
Interoperability	Documented REST/JSON API	Medium
Data Integrity	$\geq 99.9\%$ valid stored records	High
Backup	Incremental every 6h; full daily; 30-day retention	High
Recovery	RTO $\leq 4h$; RPO $\leq 1h$	High
Response Time	$\leq 2s$ for 95% of requests ($\leq 1,000$ users)	High
Compliance	Regional data protection compliance	High
Accessibility	WCAG 2.1 Level AA (web)	Medium

7. Actors

7.1 Primary Actors

Primary actors are those who directly initiate interactions with the AgriTwin system to accomplish a goal.

Primary Actor	Responsibilities
Farmer	Registers fields, views Digital Twin dashboard and predictions, receives alerts, applies recommendations, requests expert consultation.
Farm Manager	Manages multiple fields, configures crop profiles, monitors consolidated dashboards, coordinates interventions.
Agricultural Expert	Reviews flagged predictions, responds to farmer consultations, issues advisory guidance.
Government Agriculture Officer	Monitors regional aggregated crop statistics, publishes advisories, exports compliance reports.
Administrator	Manages user accounts, system configuration, monitors logs, oversees platform operation.

7.2 Secondary Actors

Secondary actors are external systems that the AgriTwin platform interacts with to obtain data or provide services, without direct human operation of the AgriTwin interface.

Secondary Actor	Responsibilities
IoT Sensor System	Automatically transmits soil, temperature, and humidity readings to the platform at scheduled intervals.
Drone System	Captures and uploads field imagery for vegetation index computation and disease detection.
Weather Service	Supplies current and forecast weather data through an external API.
Satellite Data Provider	Supplies periodic satellite imagery and spectral index data for registered field boundaries.
AI Prediction Engine	Internal automated subsystem that processes synchronized data to generate predictions consumed by other components.

8. Use Cases

This section documents eighteen (18) detailed use cases derived from stakeholder interactions with the AgriTwin platform, exceeding the minimum requirement of fifteen. Each use case specifies its actor, description, preconditions, main and alternative flows, and post-conditions.

UC-01 – Register and Login

Field	Detail
Actor	Farmer / Farm Manager / Agricultural Expert
Description	Allows a new user to create an account and existing users to authenticate into the system.
Precondition	User has a valid email address and internet access.
Main Flow	1. User opens the app. 2. User selects Register or Login. 3. User enters required details. 4. System validates input. 5. System creates account or authenticates session. 6. User is directed to role-based dashboard.
Alternative Flow	If entered credentials are invalid, the system displays an error message and allows retry up to 5 attempts before temporary lockout.
Post Condition	User account is created or an authenticated session is established.

UC-02 – Register a New Field

Field	Detail
Actor	Farmer / Farm Manager
Description	Allows a user to add a new field to the platform by defining its boundary and basic attributes.
Precondition	User is authenticated and logged in.
Main Flow	1. User selects "Add Field". 2. User draws or uploads field boundary on the map. 3. User enters field name, area, and soil type. 4. System creates the field record and initializes a Digital Twin instance.
Alternative Flow	If GPS boundary data is unavailable, the user may manually enter approximate coordinates, which the system flags for later verification.
Post Condition	A new field is registered and linked to a newly created Digital Twin model.

UC-03 – Configure Crop Profile

Field	Detail
Actor	Farmer / Farm Manager

Field	Detail
Description	Allows a user to assign crop type, variety, and planting date to a registered field.
Precondition	The field has already been registered in the system.
Main Flow	1. User selects a field. 2. User selects "Configure Crop". 3. User enters crop name, variety, and planting date. 4. System stores the crop profile and calculates expected growth stages.
Alternative Flow	If the crop variety is not in the system database, the user may submit a request to add it, pending administrator approval.
Post Condition	Crop profile is associated with the field and used for subsequent predictions.

UC-04 – View Digital Twin Dashboard

Field	Detail
Actor	Farmer / Farm Manager
Description	Allows a user to view the current synchronized state of a field, including sensor data, imagery, and health indicators.
Precondition	The field has an active Digital Twin model with at least one data synchronization completed.
Main Flow	1. User selects a field from their list. 2. System retrieves the latest synchronized Digital Twin state. 3. Dashboard renders field map, sensor readings, and health summary.
Alternative Flow	If synchronized data is more than 24 hours old, the system displays a "stale data" warning banner.
Post Condition	User views an up-to-date visual summary of the field.

UC-05 – View Crop Health Prediction

Field	Detail
Actor	Farmer / Farm Manager / Agricultural Expert
Description	Allows a user to view the AI-generated crop health status and confidence score for a field.
Precondition	At least one prediction cycle has completed for the selected field.
Main Flow	1. User navigates to the Predictions tab. 2. System displays the latest crop health status, contributing factors, and confidence score.
Alternative Flow	If confidence score is below a defined threshold, the system flags the prediction as "Requires Expert Review".

Field	Detail
Post Condition	User is informed of the current predicted crop health status.

UC-06 – Receive Alert Notification

Field	Detail
Actor	Farmer / Farm Manager
Description	Automatically notifies a user when the system detects a critical field condition.
Precondition	An alert-triggering condition has been detected by the AI Prediction Engine.
Main Flow	1. AI Prediction Engine generates a high-severity prediction. 2. System creates an alert record. 3. System sends a push notification (and SMS/email if configured) to the relevant user. 4. User opens the alert to view details.
Alternative Flow	If the user has disabled push notifications, the alert is still visible within the in-app notification centre.
Post Condition	User is informed of the critical condition and can take corrective action.

UC-07 – Request Irrigation Recommendation

Field	Detail
Actor	Farmer / Farm Manager
Description	Allows a user to request an AI-generated irrigation recommendation for a selected field.
Precondition	The field has recent soil moisture data and weather forecast data available.
Main Flow	1. User selects "Get Irrigation Recommendation". 2. System retrieves latest soil moisture and forecast data. 3. Recommendation Engine calculates optimal irrigation schedule and volume. 4. System displays the recommendation.
Alternative Flow	If soil moisture data is unavailable, the system informs the user and suggests scheduling a manual field check.
Post Condition	User receives an actionable irrigation recommendation.

UC-08 – Request Fertilization Recommendation

Field	Detail
Actor	Farmer / Farm Manager

Field	Detail
Description	Allows a user to request a fertilization plan based on detected nutrient deficiency indicators.
Precondition	Nutrient-related sensor or imagery data is available for the field.
Main Flow	1. User selects "Get Fertilization Recommendation". 2. System analyses nutrient indicators and crop growth stage. 3. Recommendation Engine generates a fertilization plan. 4. System displays the plan to the user.
Alternative Flow	If nutrient data is insufficient, the system prompts the user to schedule a soil test and provides a generic guideline instead.
Post Condition	User receives a fertilization plan tailored to current field conditions.

UC-09 – View Pest Outbreak Prediction

Field	Detail
Actor	Farmer / Farm Manager / Government Officer
Description	Allows a user to view predicted pest outbreak risk for a field or region.
Precondition	Pest prediction data has been generated for the relevant field(s).
Main Flow	1. User navigates to the Pest Risk tab. 2. System displays predicted risk level, affected pest type, and recommended monitoring window.
Alternative Flow	If risk level is "Severe", the system automatically escalates the case by generating a high-priority alert.
Post Condition	User is informed of pest outbreak risk and monitoring guidance.

UC-10 – Run Field Simulation

Field	Detail
Actor	Farmer / Farm Manager / Agricultural Expert
Description	Allows a user to simulate the outcome of a proposed intervention using the Digital Twin model.
Precondition	The field has an active, synchronized Digital Twin model.
Main Flow	1. User selects "Run Simulation". 2. User defines proposed changes (e.g., irrigation increase). 3. Simulation Module computes the projected outcome using the Digital Twin state. 4. System displays the simulated result.
Alternative Flow	If the requested simulation parameters are outside the supported range, the system displays a validation error and suggests valid bounds.
Post Condition	User obtains a projected outcome to inform a real-world decision.

UC-11 – Generate Predictive Report

Field	Detail
Actor	Farmer / Farm Manager / Agricultural Expert
Description	Allows a user to generate a downloadable report summarizing field health, risks, and recommendations for a chosen period.
Precondition	The field has sufficient historical data for the requested reporting period.
Main Flow	1. User selects "Generate Report". 2. User specifies the reporting period and report type. 3. Reporting Service compiles data and predictions. 4. System generates and makes available a downloadable PDF report.
Alternative Flow	If insufficient data exists for the selected period, the system generates a partial report and notes the data gap.
Post Condition	A predictive report is generated and available for download.

UC-12 – Request Expert Consultation

Field	Detail
Actor	Farmer
Description	Allows a farmer to submit a query to an agricultural expert regarding field conditions.
Precondition	The farmer is authenticated and has at least one registered field.
Main Flow	1. Farmer selects "Consult Expert". 2. Farmer enters query text and optionally attaches field images. 3. System routes the query to an available expert. 4. Expert receives notification of the new consultation request.
Alternative Flow	If no expert is currently available, the system queues the request and notifies the farmer of expected response time.
Post Condition	A consultation request is created and routed for expert response.

UC-13 – Respond to Consultation

Field	Detail
Actor	Agricultural Expert
Description	Allows an agricultural expert to view and respond to a farmer consultation request.
Precondition	A consultation request has been assigned to the expert.
Main Flow	1. Expert opens the consultation queue. 2. Expert reviews farmer query and associated field data. 3. Expert submits a written response. 4. System notifies the farmer that a response is available.
Alternative Flow	If the expert requires more information, the expert can request additional details from the farmer before responding fully.

Field	Detail
Post Condition	The consultation is updated with an expert response and the farmer is notified.

UC-14 – View Regional Crop Statistics

Field	Detail
Actor	Government Agriculture Officer
Description	Allows a government officer to view aggregated, anonymized crop health statistics for a selected region.
Precondition	The officer has an authenticated account with regional access rights.
Main Flow	1. Officer selects a region and date range. 2. System aggregates anonymized field data within the region. 3. System displays regional statistics and trend charts.
Alternative Flow	If insufficient aggregated data exists for statistical validity, the system displays a data-availability notice instead of unreliable figures.
Post Condition	Officer obtains a regional overview of crop health for policy planning.

UC-15 – Publish Regional Advisory

Field	Detail
Actor	Government Agriculture Officer
Description	Allows a government officer to publish an advisory message visible to farmers within an affected region.
Precondition	The officer is authenticated with advisory-publishing permission.
Main Flow	1. Officer selects "Publish Advisory". 2. Officer enters advisory text, affected region, and validity period. 3. System publishes the advisory. 4. Farmers within the region receive a notification.
Alternative Flow	If the advisory text fails content validation (e.g., missing required fields), the system prompts the officer to correct it before publishing.
Post Condition	The advisory is published and visible to affected farmers.

UC-16 – Manage User Accounts

Field	Detail
Actor	Administrator
Description	Allows an administrator to create, modify, deactivate, or delete user accounts and assign roles.

Field	Detail
Precondition	The administrator is authenticated with administrative privileges.
Main Flow	1. Administrator navigates to User Management. 2. Administrator selects a user or creates a new one. 3. Administrator modifies role or status. 4. System applies and logs the change.
Alternative Flow	If the administrator attempts to delete an account with active field data, the system requires confirmation and offers to archive instead of permanently deleting.
Post Condition	User account records are updated accordingly.

UC-17 – Monitor System Logs

Field	Detail
Actor	Administrator
Description	Allows an administrator to review system activity logs for data ingestion, prediction jobs, and user actions.
Precondition	The administrator is authenticated with administrative privileges.
Main Flow	1. Administrator navigates to System Logs. 2. Administrator applies filters (date range, module, severity). 3. System displays matching log entries.
Alternative Flow	If no log entries match the filter criteria, the system displays an empty-state message with the applied filter summary.
Post Condition	Administrator reviews relevant system activity for auditing or troubleshooting.

UC-18 – Ingest Automated Sensor Data

Field	Detail
Actor	IoT Sensor System
Description	Automated process by which the platform receives and stores periodic sensor readings.
Precondition	The sensor is registered and associated with a field.
Main Flow	1. Sensor transmits a reading to the IoT gateway. 2. System validates the reading format and range. 3. System stores the reading and updates the field's current state. 4. System triggers Digital Twin synchronization if thresholds are met.
Alternative Flow	If a reading fails validation, the system logs the anomaly and excludes it from the Digital Twin update, optionally alerting the administrator if failures persist.
Post Condition	A validated sensor reading is stored and reflected in the field state.

9. UML Use Case Diagram

The following UML Use Case Diagram illustrates the primary and secondary actors of the AgriTwin system and their associations with the major use cases described in Section 8. The diagram groups use cases within the system boundary of the AgriTwin Digital Twin Platform and shows representative associations between each actor and the use cases they participate in.

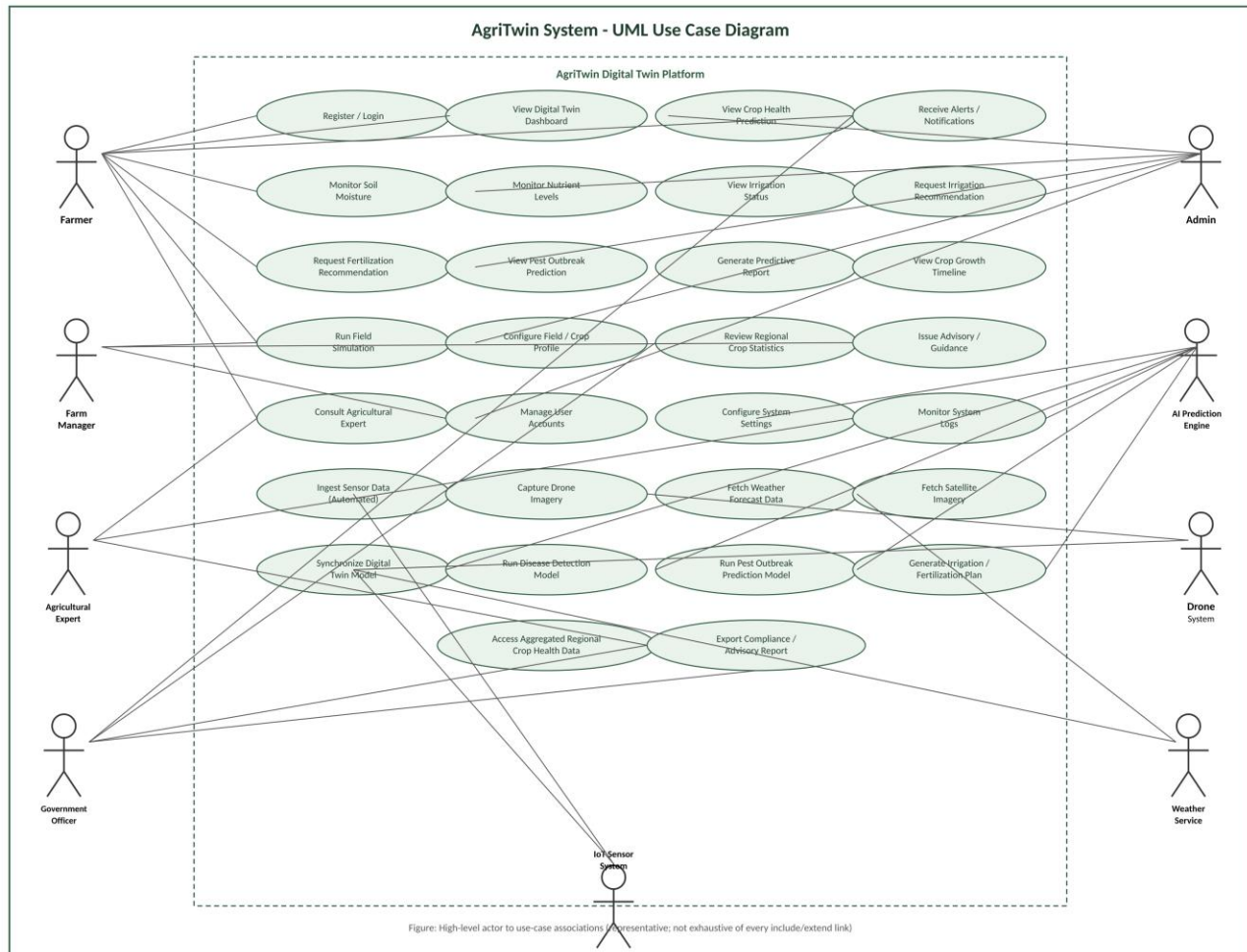


Figure 9.1: UML Use Case Diagram for the AgriTwin Digital Twin Platform

An editable vector source for this diagram (SVG) is provided in Appendix A for recreation or modification in any diagramming tool that supports SVG import, such as Microsoft Visio, draw.io, or Lucidchart.

10. Data Requirements

This section presents the complete data analysis for the AgriTwin platform, including identified entities, their attributes, primary and foreign keys, and the relationships between them. The corresponding Entity Relationship Diagram is presented in Figure 10.1.

10.1 Entities and Attributes

Entity	Key Attributes	Description
USER	PK user_id; full_name; role; email; phone_number; password_hash; created_at	Represents any registered human user of the system (Farmer, Farm Manager, Expert, Officer, or Admin).
FARM	PK farm_id; FK owner_user_id; farm_name; location_gps; total_area_ha; soil_type	Represents a farm property owned or operated by a user, which may contain multiple fields.
FIELD	PK field_id; FK farm_id; field_name; boundary_polygon; area_ha; current_crop_id; digital_twin_id	Represents an individually managed field within a farm, the core monitored unit.
DIGITAL_TWIN_MODEL	PK twin_id; FK field_id; model_version; last_sync_time; sync_status; simulation_state	Represents the virtual model instance synchronized with a physical field.
CROP	PK crop_id; crop_name; variety; growth_duration_days; ideal_moisture_pct; ideal_temp_range	Reference entity describing crop types and their ideal growing parameters.
SENSOR	PK sensor_id; FK field_id; sensor_type; install_date; status; battery_level	Represents a physical IoT sensor device deployed in a field.
SENSOR_READING	PK reading_id; FK sensor_id; parameter_name; value; unit; recorded_at	Represents an individual data point captured by a sensor.
WEATHER_DATA	PK weather_id; FK field_id; temperature; humidity; rainfall_mm; wind_speed; forecast_date	Represents weather observations or forecasts for a field's location.
DRONE_IMAGE	PK image_id; FK field_id; drone_id; capture_time; image_url; resolution; ndvi_index	Represents an image captured by a drone over a field.
SATELLITE_IMAGE	PK sat_image_id; FK field_id; provider_name; capture_date; image_url; spectral_band	Represents satellite imagery covering a field.

Entity	Key Attributes	Description
PREDICTION_RESULT	PK prediction_id; FK field_id; FK twin_id; prediction type; result_label; confidence_score; generated_at	Represents an AI-generated prediction (health, disease, or pest) for a field.
RECOMMENDATION	PK recommendation_id; FK prediction_id; FK field_id; category; description; priority_level; status	Represents an actionable recommendation generated from a prediction result.
ALERT	PK alert_id; FK field_id; FK user_id; alert_type; severity; message; is_read	Represents a notification triggered by a critical prediction or condition.
REPORT	PK report_id; FK field_id; FK generated_by; report_type; file_url; period_start; period_end	Represents a generated predictive or historical report document.
IRRIGATION_LOG	PK irrigation_id; FK field_id; FK recommendation_id; water_volume_l; method; performed_at	Represents a record of an irrigation action performed, optionally linked to a recommendation.
EXPERT_CONSULTATION	PK consultation_id; FK field_id; FK farmer_user_id; FK expert_user_id; query_text; response_text; status	Represents a consultation exchange between a farmer and an agricultural expert.

10.2 Relationships

- One USER (owner) has many FARM records (1:N).
- One FARM contains many FIELD records (1:N).
- One FIELD has exactly one DIGITAL_TWIN_MODEL (1:1).
- One CROP type may be assigned to many FIELD records (1:N).
- One FIELD is monitored by many SENSOR devices (1:N).
- One SENSOR produces many SENSOR_READING records (1:N).
- One FIELD accumulates many WEATHER_DATA, DRONE_IMAGE, and SATELLITE_IMAGE records over time (1:N each).
- One DIGITAL_TWIN_MODEL and FIELD combination produces many PREDICTION_RESULT records over time (1:N).
- One PREDICTION_RESULT may generate one or more RECOMMENDATION records (1:N).
- One RECOMMENDATION or PREDICTION_RESULT may trigger one ALERT (1:1 or 1:N depending on severity).

- One FIELD has many REPORT records generated over time (1:N).
- One RECOMMENDATION may be linked to zero or one IRRIGATION_LOG entry recording its execution (1:0..1).
- One FARMER (USER) and one EXPERT (USER) participate in many EXPERT_CONSULTATION records (M:N resolved via the consultation entity).

10.3 Entity Relationship Diagram

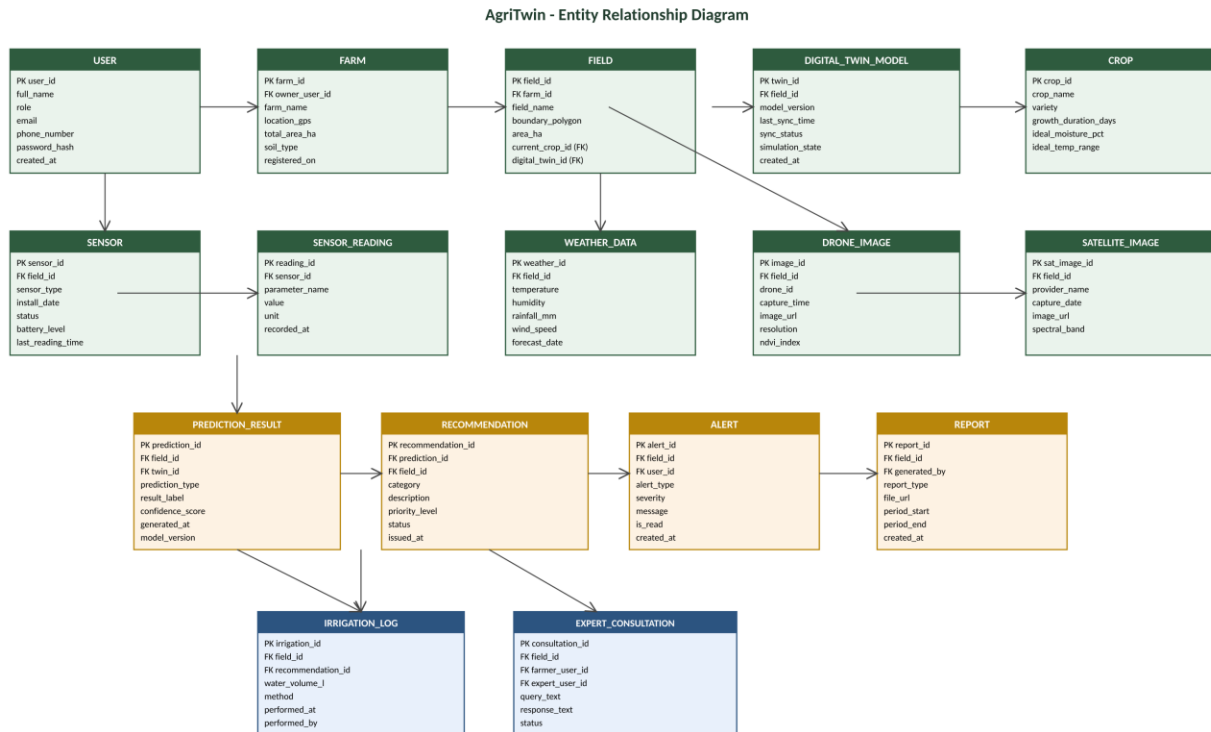


Figure: AgriTwin ERD — crow's-foot notation; PK = Primary Key, FK = Foreign Key

Figure 10.1: Entity Relationship Diagram for the AgriTwin database (crow's-foot notation)

11. External Interface Requirements

11.1 User Interface

The system shall provide a responsive web application and a native/hybrid mobile application, both following a consistent visual design language. Dashboards shall use map-based field visualization, colour-coded health indicators (green/amber/red), and card-based summaries for alerts and recommendations. All interactive elements shall provide immediate visual feedback upon user action.

11.2 Hardware Interface

The system shall interface with IoT soil and climate sensors via LoRaWAN or cellular (4G/NB-IoT) gateways that forward data using standard messaging protocols. The system shall interface with drone onboard controllers to receive uploaded imagery either via direct cellular upload or through a ground-station relay after the drone lands.

11.3 Software Interface

The system shall interface with a relational database management system (such as PostgreSQL) for structured data and an object storage service for imagery and report files. The AI Prediction Engine shall interface with the Digital Twin Engine through internal service APIs using JSON payloads.

11.4 Communication Interface

All client-server communication shall occur over HTTPS using TLS 1.2 or higher. IoT sensor communication shall use MQTT over TLS where supported by the device gateway. Asynchronous internal processing shall use a message queue (such as Kafka or RabbitMQ) for event-driven communication between ingestion, prediction, and notification services.

11.5 API Interface

The system shall expose a versioned, documented REST API (e.g., /api/v1/) supporting JSON request and response payloads, token-based authentication (OAuth 2.0 or JWT), and standard HTTP status codes for error handling, enabling both internal client applications and authorized third-party integrations.

11.6 Cloud Services

The system shall be deployed on a cloud infrastructure provider supporting container orchestration (such as Kubernetes), managed relational database services, object

storage, and managed message queue services, to satisfy the scalability and availability requirements defined in Section 6.

12. System Features

This section describes, in detail, the sixteen (16) major features of the AgriTwin platform, exceeding the minimum requirement of fifteen.

12.1 Digital Twin Field Modelling

AgriTwin maintains a continuously updated virtual replica of each registered field, integrating boundary geometry, soil characteristics, crop profile, and the latest synchronized sensor, imagery, and weather data into a single coherent model that reflects the real-world state of the field.

12.2 Multi-Source Real-Time Data Integration

The platform ingests data from IoT sensors, drones, weather stations, and satellite imagery providers through a unified ingestion layer, normalizing heterogeneous formats and timestamps into a consistent internal data model.

12.3 AI-Driven Crop Health Prediction

Machine learning models analyse synchronized field data to classify crop health status (healthy, stressed, critical) and provide a confidence score, enabling early identification of issues before they are visually apparent.

12.4 Disease Detection from Imagery

Computer vision models process drone and satellite imagery to detect visual patterns associated with common crop diseases, flagging affected zones within the field for targeted inspection.

12.5 Pest Outbreak Forecasting

Predictive models combine weather patterns, historical pest incidence data, and current field conditions to forecast the likelihood and severity of pest outbreaks, allowing preventive action.

12.6 Intelligent Irrigation Recommendations

The Recommendation Engine calculates optimal irrigation timing and volume based on real-time soil moisture, crop water requirements, and short-term weather forecasts, helping to reduce water wastage.

12.7 Fertilization Guidance

The system generates nutrient application recommendations aligned with the crop's current growth stage and any detected nutrient deficiency indicators, improving fertilizer use efficiency.

12.8 Pest Management Recommendations

When pest outbreak risk exceeds a configurable threshold, the system recommends appropriate management actions, ranging from monitoring intensification to specific control measures.

12.9 Real-Time Alerting and Notifications

Critical conditions automatically trigger alerts delivered through in-app notifications, and optionally SMS or email, ensuring timely awareness even when the user is not actively using the application.

12.10 Visualization Dashboards

Role-specific dashboards present field maps, sensor trend charts, health indicators, and recommendation summaries in an intuitive, colour-coded format suited to each stakeholder's needs.

12.11 Crop Growth Timeline Monitoring

The platform tracks and visualizes crop growth stages over the full crop cycle, correlating each stage with historical health and environmental data for retrospective analysis.

12.12 Field Simulation ("What-If" Analysis)

Users can simulate the projected impact of a proposed intervention, such as an altered irrigation schedule, on the Digital Twin model before applying it physically in the field.

12.13 Predictive and Historical Reporting

The system generates structured, downloadable reports summarizing field health trends, risk events, and recommendations over a selected period, suitable for record-keeping or sharing with stakeholders.

12.14 Expert Consultation Workflow

A structured consultation workflow connects farmers with agricultural experts, allowing queries with attached field data to be routed, reviewed, and answered within the platform.

12.15 Regional Aggregated Analytics

Government Agriculture Officers can access anonymized, aggregated crop health statistics at a regional level, supporting policy planning and early public advisory issuance.

12.16 Role-Based Access Control and Administration

The Administrator role provides centralized management of user accounts, permissions, and system configuration, along with visibility into system logs for operational oversight.

13. Assumptions

The following assumptions have been made in the preparation of this SRS and the design of the AgriTwin platform:

- Registered fields have reliable internet connectivity, either directly or through an intermediate gateway device, sufficient to transmit sensor and imagery data at the required frequency.
- Farmers and farm managers possess a basic level of smartphone literacy sufficient to navigate a modern mobile application.
- IoT sensors deployed in the field are properly calibrated and maintained by the farmer or a designated technician.
- Third-party weather and satellite data providers maintain their published API availability and data accuracy commitments.
- Drone operations comply with applicable regional aviation and airspace regulations, and drone operators are appropriately licensed where required.
- Agricultural experts allocated to the platform have sufficient domain expertise to validate AI-generated predictions and respond to farmer queries.
- The initial release targets a defined set of common crop varieties, with model accuracy expected to be highest for these supported crops.
- Users have access to a device (smartphone, tablet, or computer) capable of running the required client application.
- Historical agricultural and weather data of sufficient volume and quality is available to train and validate the initial AI/ML models.
- Government agencies accessing regional statistics operate under established data-sharing agreements that permit access to anonymized, aggregated farm data.
- Cloud infrastructure services required for hosting (compute, storage, messaging) are available in the deployment region with acceptable latency to end users.

14. Constraints

Constraint Category	Description
Technical	The system must be built using technologies compatible with cloud-native, containerized deployment and must support integration with heterogeneous IoT device protocols (LoRaWAN, MQTT, cellular).
Operational	The platform must remain operational with degraded (not failed) functionality when one or more external data sources (weather, satellite) are temporarily unavailable.
Economic	Development and operational costs must remain within the budget allocated for an academic-scale pilot deployment, favouring open-source and managed cloud services over costly proprietary licenses where feasible.
Legal	The system must comply with applicable data protection and privacy regulations governing the collection, storage, and processing of farmer location and personal data.
Environmental	Field-deployed IoT hardware must be capable of operating reliably under outdoor environmental conditions, including temperature extremes, humidity, and dust exposure.
Time	The system must be delivered within the academic project timeline defined for the CSA1012 Software Engineering course assignment.
Hardware	The system depends on the availability and proper functioning of externally sourced IoT sensors, drones, and their associated firmware, which are outside direct control of the software development team.
Software	AI/ML model performance is constrained by the availability and quality of labelled training data for crop disease and pest classification.
Network	Field connectivity in remote agricultural areas may be limited or intermittent, requiring the system to tolerate delayed or batched data transmission from field devices.
Budget	The scope of AI model sophistication and the number of integrated third-party data providers is constrained by the available project budget for API subscription costs.

15. Requirement Validation

This section documents the validation process applied to the requirements captured in this SRS, evaluating them against the criteria of completeness, consistency, correctness, traceability, ambiguity, and redundancy, and identifies areas for improvement.

15.1 Completeness

The requirement set was reviewed against the assignment scope and the objectives defined in Section 2.4 to confirm that every stated objective (Digital Twin modelling, multi-source data collection, AI-based prediction, monitoring, recommendation, visualization, and reporting) is addressed by at least one functional requirement and one corresponding use case. All sixteen system features map to one or more functional requirements, and no orphaned objectives were identified.

15.2 Consistency

Requirements were cross-checked to ensure that no two requirements impose contradictory behaviour on the same system function. For example, the alerting requirements (FR-16, FR-17) consistently reference the same alert severity model used in the Non-Functional Requirements (Section 6.1) and the ALERT entity defined in Section 10.1.

15.3 Correctness

Each functional requirement was reviewed against the original problem statement and objectives to confirm that it accurately reflects a genuine need of the AgriTwin scenario, rather than an invented or unrelated feature. Requirements involving external systems (IoT, drone, weather, satellite) were verified against the stakeholder responsibilities defined in Section 3.

15.4 Traceability

A traceability check was performed to confirm that every functional requirement can be traced to at least one stakeholder need and at least one use case, and that every use case can be traced back to a specific objective. Representative traceability is summarized in the checklist table below.

15.5 Ambiguity

Requirements were reviewed for vague or subjective terms (such as "fast" or "user-friendly") and, where such qualities were required, they were replaced with measurable targets, as reflected in the Non-Functional Requirements in Section 6 (for example, specific response time and uptime figures rather than general statements).

15.6 Redundancy

The requirement set was checked for duplicate or overlapping requirements. Related but distinct requirements, such as FR-13 (Irrigation Recommendation) and FR-14 (Fertilization Recommendation), were confirmed to address separate recommendation categories rather than representing redundant statements of the same requirement.

15.7 Missing Requirements

During validation, an additional requirement for multi-field management (FR-29) was identified as necessary to fully support the Farm Manager stakeholder role and was incorporated into the final requirement set to close this gap.

15.8 Suggested Improvements

- Future iterations should define explicit data retention periods for raw sensor readings versus aggregated/derived data to optimize storage costs.
- A formal model-accuracy Service Level Agreement (SLA) should be defined for the AI Prediction Engine as historical accuracy data becomes available post-deployment.
- Offline/low-connectivity operation modes for the mobile application should be further detailed in a subsequent revision to support farmers in low-network areas.

15.9 Validation Checklist

Validation Criterion	Status	Remarks
Completeness	Passed	All stated objectives map to functional requirements and use cases.
Consistency	Passed	No contradictory requirements identified across sections.
Correctness	Passed	All requirements verified against the original problem statement.
Traceability	Passed	Requirements traced to stakeholders, use cases, and features.
Ambiguity	Passed	Subjective terms replaced with measurable NFR targets.
Redundancy	Passed	No duplicate requirements found; related items are distinct.
Missing Requirements	Resolved	FR-29 added to address Farm Manager multi-field need.

Validation Criterion	Status	Remarks
Feasibility	Passed	Requirements are achievable within stated technical and budget constraints.

16. Conclusion

This Software Requirement Specification has presented a complete and structured analysis of the AgriTwin Digital Twin Platform for Precision Farming and Crop Health Prediction, developed in response to the growing need for integrated, intelligent, and proactive farm management solutions. Beginning with an analysis of the current manual and fragmented approach to farm monitoring, this document has identified the key stakeholders, defined the boundaries of the proposed system, and specified twenty-nine functional requirements and a comprehensive set of measurable non-functional requirements spanning performance, security, reliability, and related quality attributes.

Eighteen detailed use cases and a supporting UML Use Case Diagram describe how each stakeholder, human and automated, interacts with the system to achieve their goals. The data requirements and accompanying Entity Relationship Diagram establish a sound structural foundation for the underlying database, while the external interface requirements and system architecture ensure that AgriTwin can integrate reliably with IoT sensors, drones, weather services, and satellite data providers.

By continuously synchronizing real-time and historical field data into a virtual Digital Twin and applying AI-driven predictive analysis, AgriTwin is positioned to enable earlier detection of crop stress, disease, and pest outbreaks, more efficient use of water and fertilizer resources, and better-informed decision-making for farmers, agricultural experts, and government agencies alike. The requirements captured in this document, having been validated for completeness, consistency, correctness, and traceability, provide a solid and unambiguous foundation for the subsequent design and implementation phases of the AgriTwin platform, in support of more productive and sustainable agricultural practices.

Appendix A: Supplementary Diagrams

This appendix provides the remaining professionally formatted diagrams required for the AgriTwin SRS, including the Context Diagram, Level-0 Data Flow Diagram, System Architecture Diagram, Component Diagram, and Deployment Diagram. Editable vector (SVG) sources for all diagrams in this document, including the UML Use Case Diagram and ERD presented earlier, are available on request for recreation in tools such as Microsoft Visio, draw.io, or Lucidchart.

A.1 Context Diagram (DFD Level 0)

The context diagram represents AgriTwin as a single process and shows the flow of information between the system and each external entity, providing a top-level, entity-relationship-free view of system boundaries.

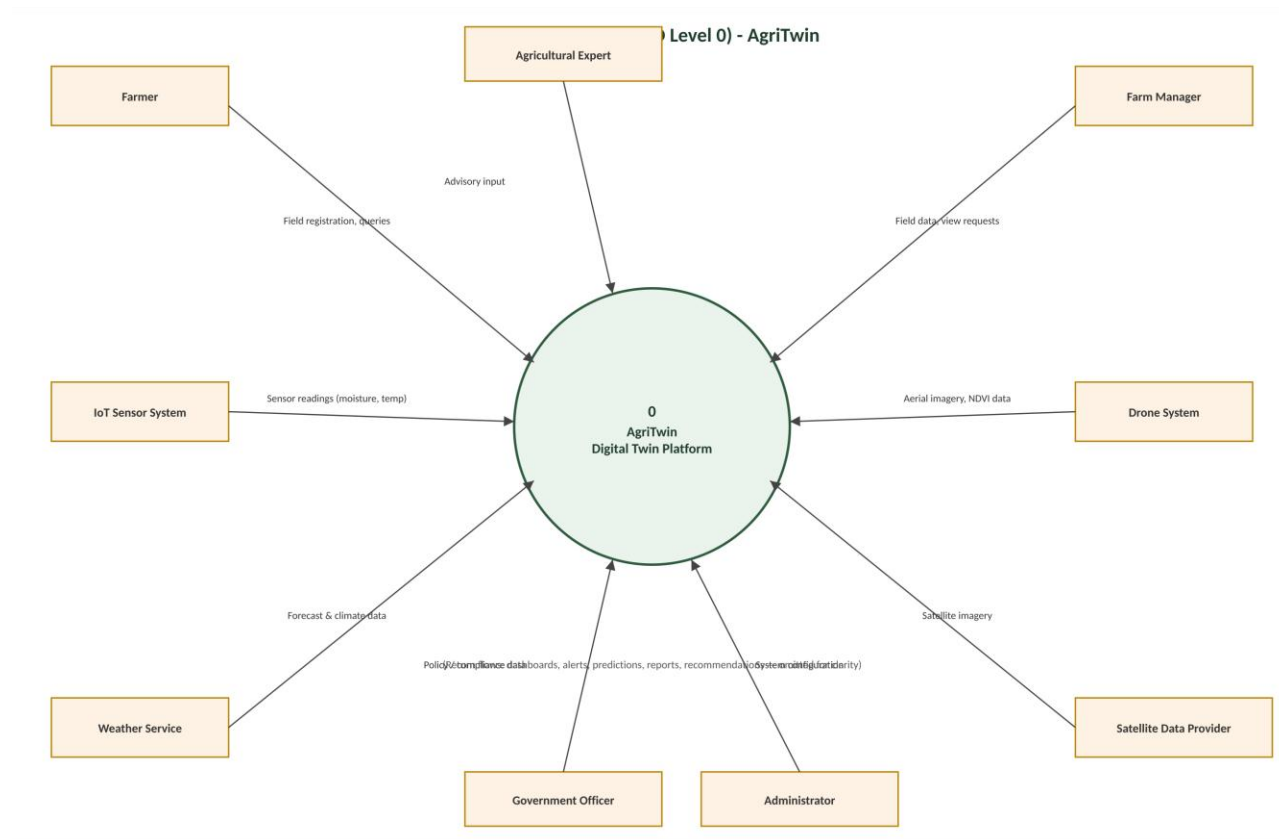


Figure A.1: Context Diagram showing AgriTwin and its external entities

A.2 Level-0 Data Flow Diagram

The Level-0 DFD decomposes the single AgriTwin process from the context diagram into its seven major internal processes and the data stores each process reads from or writes to.

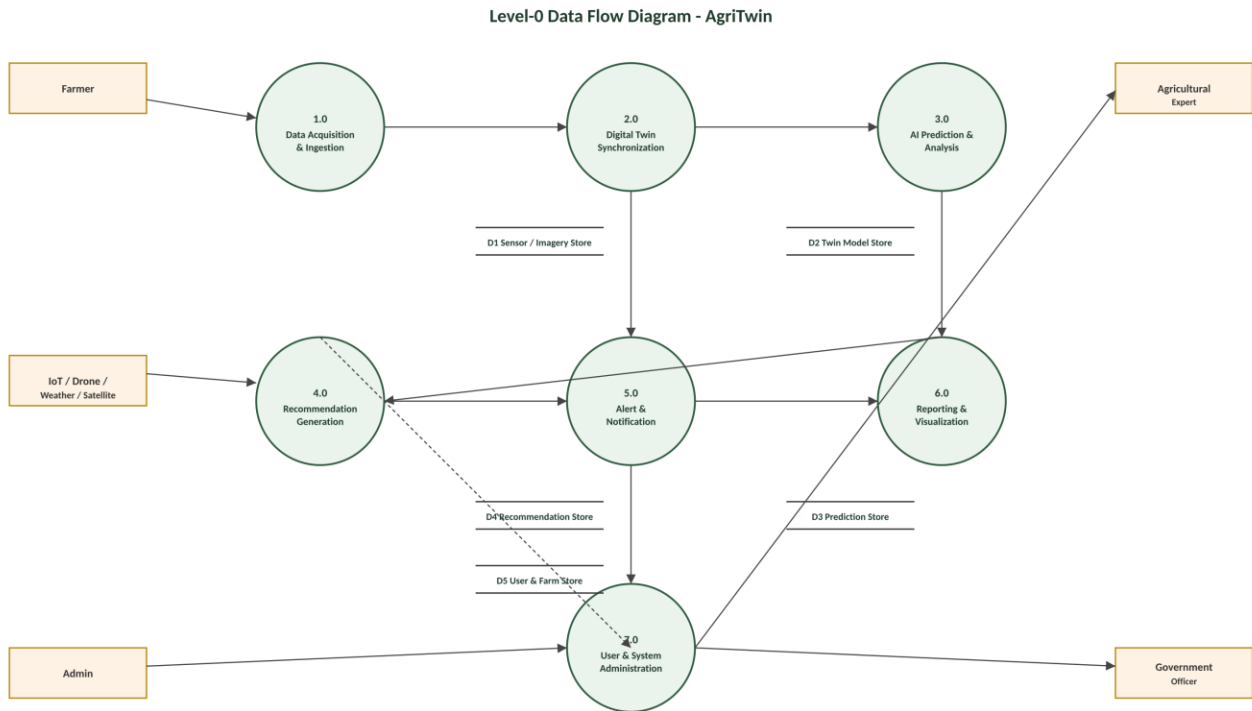


Figure: Level-0 DFD showing seven major processes and five persistent data stores

Figure A.2: Level-0 DFD decomposing AgriTwin into seven major processes

A.3 System Architecture Diagram

The system architecture diagram presents the four-layer architecture of AgriTwin, spanning field data sources, the data ingestion and integration layer, the core application and AI platform, and the presentation layer.

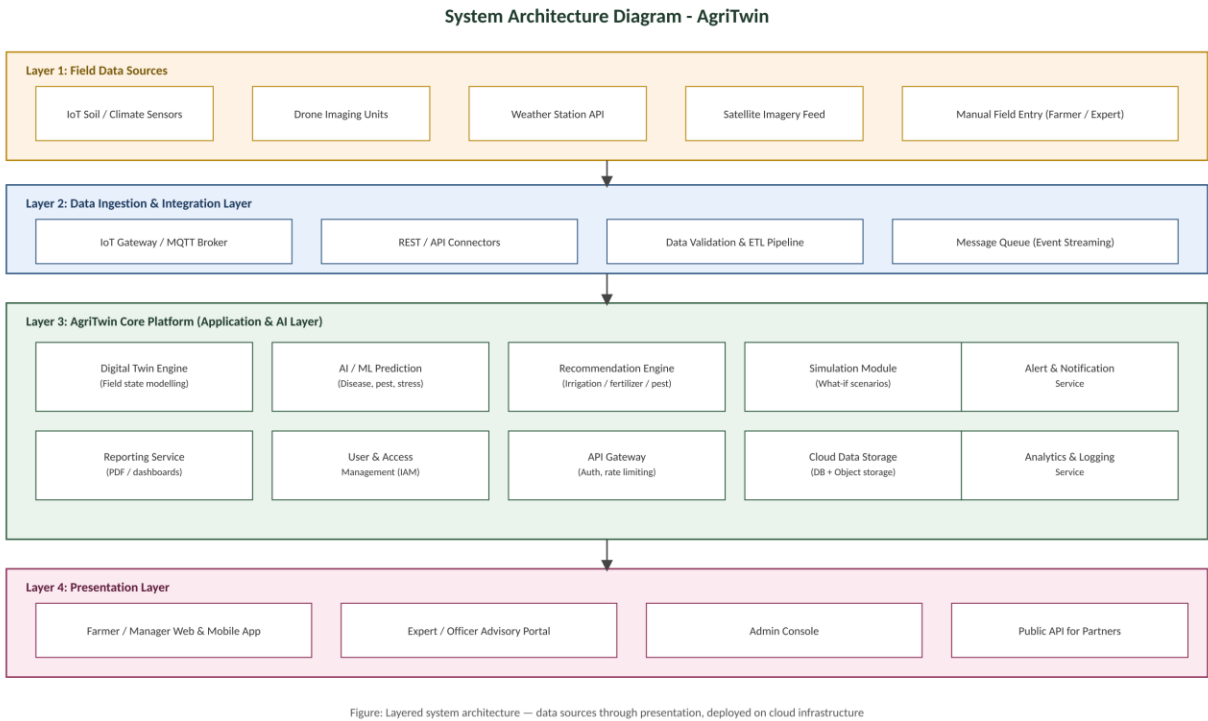


Figure A.3: Layered system architecture of the AgriTwin platform

A.4 Component Diagram

The component diagram illustrates the internal software components of the AgriTwin platform and their dependency relationships, supporting the modular, service-oriented architecture described in Section 6.6.

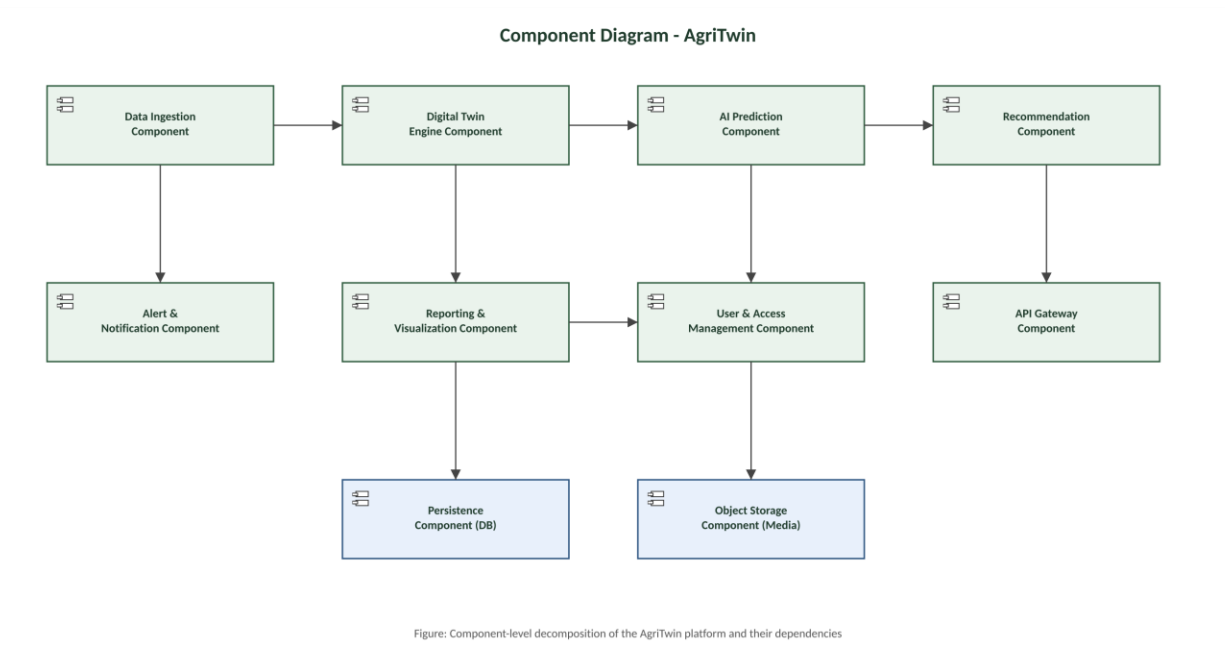


Figure A.4: Component diagram showing major software components and dependencies

A.5 Deployment Diagram

The deployment diagram shows the physical distribution of AgriTwin components across field devices, external service providers, the cloud application server cluster, and client devices.

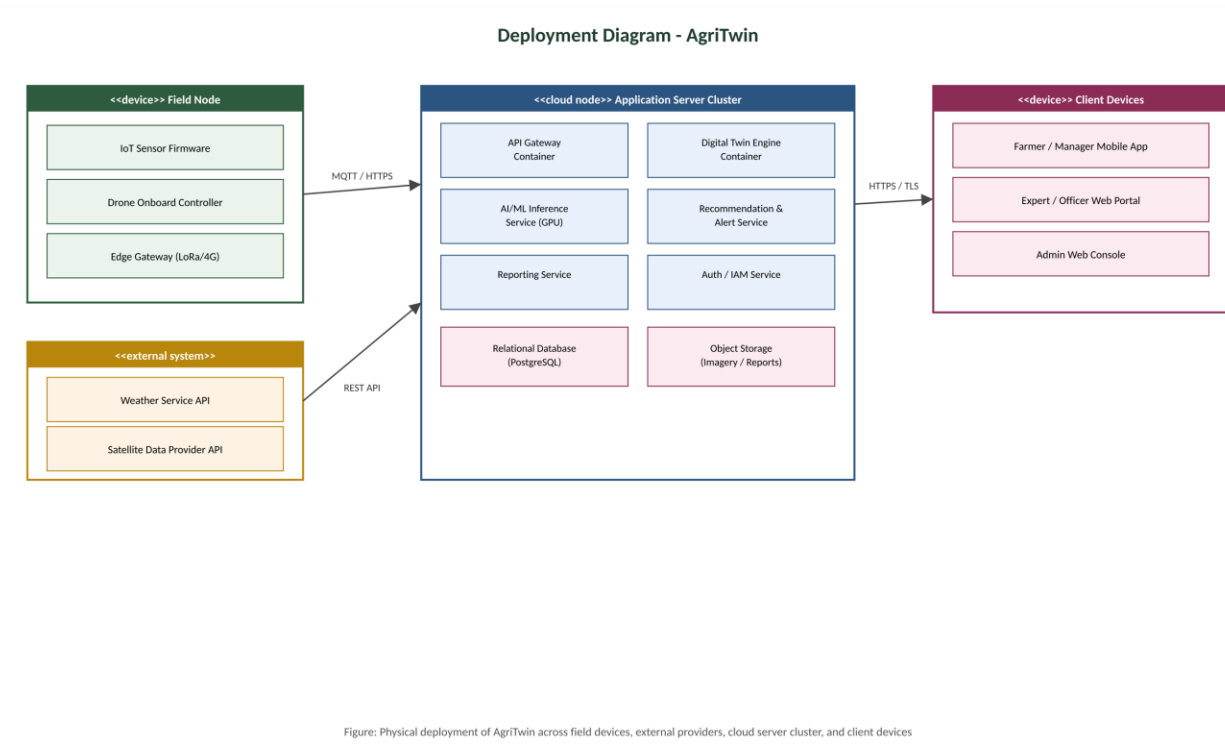


Figure A.5: Deployment diagram showing physical distribution of AgriTwin components